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import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer, WordNetLemmatizer
from nltk.probability import FreqDist

# Run these only once
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('averaged_perceptron_tagger')

# ----- User Input -----
resume = input("Enter the resume text:\n")

# ----- 1. Tokenization -----
tokens = word_tokenize(resume)
print("\n1. Tokens:")
print(tokens)

# ----- 2. Remove Stopwords -----
stop_words = set(stopwords.words('english'))

filtered_words = [
    word.lower()
    for word in tokens
    if word.isalpha() and word.lower() not in stop_words
]

print("\n2. After Removing Stopwords:")
print(filtered_words)

# ----- 3. Stemming -----
ps = PorterStemmer()
stemmed_words = [ps.stem(word) for word in filtered_words]

print("\n3. Stemmed Words:")
print(stemmed_words)

# ----- 4. Lemmatization -----
lemmatizer = WordNetLemmatizer()
lemmatized_words = [lemmatizer.lemmatize(word) for word in filtered_words]

print("\n4. Lemmatized Words:")

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print(lemmatized_words)

# ----- 5. POS Tagging -----
pos_tags = nltk.pos_tag(filtered_words)

print("\n5. POS Tags:")
for word, tag in pos_tags:
    print(word, ":", tag)

# ----- 6. Frequency Distribution -----
fdist = FreqDist(filtered_words)

print("\n6. Frequency Distribution:")
for word, freq in fdist.items():
    print(word, ":", freq)
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